

Volume and surface integrals. Gauss's and Stokes's theorems.

[SUBMIT ANSWERS TO ALL QUESTIONS]

Volume integral: $\int_V f dV$ or $\iiint_V f(x, y, z) dx dy dz$ (in Cartesian coordinates).

Surface integral of the first kind $\int_S f dS$; surface integral of the second kind $\int_S \mathbf{A} \cdot d\mathbf{S}$, where $d\mathbf{S} = \mathbf{n} dS$ and $\mathbf{n}$ is a unit vector normal to the surface.

For a surface in parametric form, $\mathbf{r} = \mathbf{r}(u, v)$, the surface element is $d\mathbf{S} = \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} du dv$.

Gauss's theorem:

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{A} dV,$$

where V is the volume enclosed by the surface S , and $d\mathbf{S}$ is outwards from V .

Stokes's theorem:

$$\oint_C \mathbf{A} \cdot d\mathbf{r} = \int_S \nabla \times \mathbf{A} \cdot d\mathbf{S},$$

where S is a surface that spans the closed curve C , and where the direction of $d\mathbf{S}$ is related to the direction in which C is traversed by the right-handed corkscrew rule.

1. (a) Evaluate $\iiint_V xyz dx dy dz$, where V is defined by $0 \leq x \leq 1$, $0 \leq y \leq x$, $0 \leq z \leq xy$. [6]

(b) Evaluate the integral $\int_0^1 dx \int_0^1 dy \int_0^1 \frac{dz}{\sqrt{x+y+z+1}}$. [7]

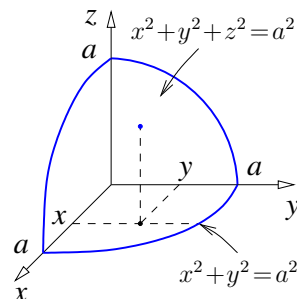
(c) Evaluate $\iiint_V e^{x+y+z} dx dy dz$, where V is a tetrahedron bounded by the planes $x + y + z = 1$, $x = 0$, $y = 0$, $z = 0$. [7]

2. Evaluate the volume integral $\int_V xyz dV$, where V is bounded by the surfaces $x^2 + y^2 + z^2 = a^2$ (a sphere of radius a), $x = 0$ (the $y - z$ plane), $y = 0$ (the $x - z$ plane), $z = 0$ (the $x - y$ plane), and located in the first octant, i.e., for $x \geq 0$, $y \geq 0$ and $z \geq 0$, (see diagram)

(a) using Cartesian coordinates;

(b) using spherical coordinates. [10]

[10]



3. Consider a vector field $\mathbf{A} = x^2 y z \mathbf{i} + x y^2 z \mathbf{j} + x y z^2 \mathbf{k}$.

(a) Evaluate $\oint_S \mathbf{A} \cdot d\mathbf{S}$ for the surface S that bounds volume V in Question 2. [10]

(b) Find $\nabla \cdot \mathbf{A}$ and obtain $\oint_S \mathbf{A} \cdot d\mathbf{S}$ from Gauss's theorem using the result of Question 2. [10]

4. (a) For the vector field $\mathbf{B} = y^2\mathbf{i} + z^2\mathbf{j} + x^2\mathbf{k}$, evaluate $\int_S \mathbf{B} \cdot d\mathbf{S}$, where S is
- i. a hemisphere, $x^2 + y^2 + z^2 = a^2$, $z \geq 0$, with $d\mathbf{S}$ along $\mathbf{r}$; [10]
 - ii. a disk, $x^2 + y^2 \leq a^2$, $z = 0$, with $d\mathbf{S}$ along $\mathbf{k}$. [10]
- (b) For $\mathbf{A} = \frac{1}{3}(z^3\mathbf{i} + x^3\mathbf{j} + y^3\mathbf{k})$, find $\oint_C \mathbf{A} \cdot d\mathbf{r}$ for the curve C : $x^2 + y^2 = a^2$, $z = 0$, traversed anticlockwise when viewed from positive z . [10]
- (c) Show that $\mathbf{B} = \nabla \times \mathbf{A}$, and use Stokes's theorem to explain the relation between the answers in parts (a) and (b). [10]